EduSkillBench: Evaluating Procedural Skill Use in Educational Agents

Anonymous ACL submission

Abstract

Reusable Skills can guide educational agents through teaching procedures, but their availability does not establish that an agent uses them effectively. We present EDUSKILLBENCH, a benchmark of 305 tasks across 14 educational Skill families, linking teaching situations, procedural resources, execution records, and artifact-level rubrics. Paired model evaluations show small, mixed aggregate changes despite expected-package loading rates of 17.70–71.48% across four native-access systems. Skill-level breakdowns and paired traces expose both useful instructional operations and failures after loading, including unavailable resources and incomplete delivery. A 28-task expert-adjudicated study covers all 14 families and finds that close reference-relative scores can conceal larger component-score disagreements. We further examine learning from 246 Skill-informed artifacts through Qwen3-4B SFT and subsequent outcome-based reinforcement learning, alongside an execution-cost pilot. EDUSKILLBENCH provides a connected evaluation of procedural availability, observed use, and usable educational output.

1 Introduction

Educational assistants must turn knowledge into instructional decisions: which hint to offer after an unsuccessful attempt, what evidence would demonstrate understanding, and how to organize practice within a limited timetable. Producing a plausible explanation of a teaching method is easier than adapting that method to a learner and delivering materials a teacher can use. This distinction is central to pedagogical content knowledge and scaffolding (Shulman, 1986; Wood et al., 1976), and remains relevant to language models: MathTutorBench finds that subject-solving ability does not automatically translate into effective tutoring (Macina et al., 2025).

Recent work addresses this gap through pedagogical instruction following and explicit expert decisions (LearnLM Team, 2024; Wang et al., 2024b). Reusable *Agent Skills* offer a complementary approach: package a procedure, its templates, and supporting resources so that an agent can consult them when needed (Agent Skills, 2026). For example, an adaptive-hint package can connect evidence of a student's difficulty to graduated assistance and a check for independent understanding. Its educational value depends on whether the agent selects applicable operations and completes them correctly. Merely making the document available does not establish that connection.

Existing benchmarks provide complementary evidence. EduBench covers diverse educational requests, while MathTutorBench and MRBench assess open-ended tutoring behavior (Xu et al., 2025; Macina et al., 2025; Maurya et al., 2025). SkillsBench evaluates procedural packages through paired agent runs and executable verification (Li et al., 2026). Open-ended teaching products introduce a further challenge: a generated lesson, diagnostic question, or practice schedule may be fluent and procedurally detailed yet infeasible or substantively incorrect. Measuring Skill use in this setting requires linking a teaching task to its procedural resource, the agent's actions, and the quality of the delivered product.

We introduce EDUSKILLBENCH, a unified benchmark of 305 tasks across 14 educational Skill families. Each task binds a teaching situation, a designated Skill, and applicable rubric criteria. Paired NO SKILL and WITH SKILL evaluations compare delivered outcomes with and without procedural support; native execution records show whether agents actually load it. Reference-conditioned judgments assess the delivered text provided to the evaluator, while a 28-task expert-adjudicated sample checks how those judgments behave across the 14 families. A complementary

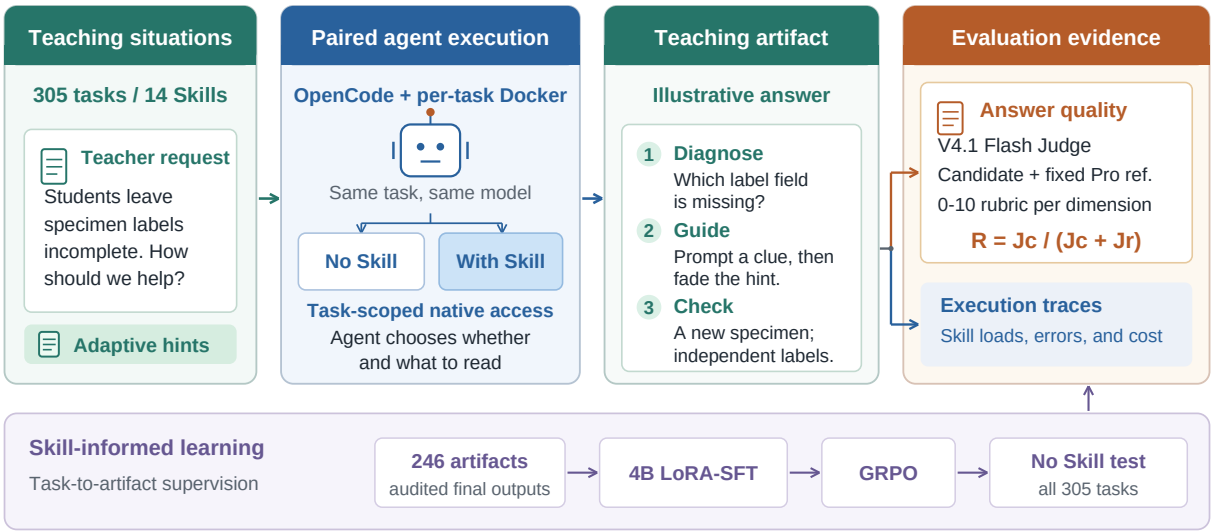


Figure 1: Overview of EDUSKILLBENCH. Teaching tasks bind procedural resources to artifact requirements and rubric criteria. Paired native execution exposes a task-matched Skill for autonomous access; a separate artifact-learning study tests performance without Skill documents. Retained outputs support outcome, access, and human–judge analyses. Rubrics and evaluation references are used after generation. Exposure details are in Appendix A.

Qwen3-4B study examines learning from Skill-informed artifacts through supervised fine-tuning (SFT) and subsequent outcome-based reinforcement learning.

The analysis reveals a gap between access and useful delivery. Across four native-access models, expected-package loading ranges from 17.70% to 71.48%, but aggregate With–No changes remain between -1.19 and $+0.50$ score points. Family breakdowns and paired cases locate more explicit teaching operations, resource failures, and incomplete deliverables behind those averages. Human-adjudicated scores further show that close reference-relative scores can coexist with substantial differences in their underlying quality components. These findings motivate evaluating procedural support at both the execution and artifact levels.

Our contributions are threefold: (1) a 305-task educational benchmark linking 14 reusable Skills to explicit output requirements and rubrics; (2) paired model evaluations with family-level outcomes, observable Skill access, and trace-based explanations; and (3) complementary studies of artifact learning, expert-adjudicated evaluation, and execution cost. Together, they make the path from procedural guidance to educational work empirically inspectable.

2 Related Work

Educational capability and teaching decisions. Educational evaluation increasingly distinguishes knowing an answer from helping a learner. MathTutorBench evaluates open-ended pedagogical abilities, and MRBench organizes the assessment of remediation responses (Macina et al., 2025; Maurya et al., 2025). EduBench extends coverage to diverse teacher- and student-facing scenarios (Xu et al., 2025); EduIllustrate evaluates the quality of instructional products across several subjects and dimensions (Bi et al., 2026). On the intervention side, Bridge models expert decisions about mistakes and remediation, LearnLM targets pedagogical instruction following, and Tutor CoPilot studies expert-inspired assistance in live tutoring (Wang et al., 2024b; LearnLM Team, 2024; Wang et al., 2025). EDUSKILLBENCH focuses on a different unit of evaluation: a teaching task linked to a reusable procedure and the agent execution that produces its instructional artifact. Artifact quality does not by itself establish student learning gains.

External Skills and learned procedures. ReAct connects actions to observations, while Voyager retains reusable programs in a Skill library (Yao et al., 2023; Wang et al., 2024a). Agent Skills provide a document-and-resource interface for procedural guidance (Agent Skills, 2026); SkillsBench tests curated packages through paired task executions (Li et al., 2026). We examine autonomous use of a task-matched educational candidate, where a correct package may be available but unread or incompletely applied. Param-

eter learning poses a separate question. SkillGym trains from successful Skill-enabled interactions, and SKILL0 studies progressive withdrawal of Skill context during reinforcement learning (Ge et al., 2026; Lu et al., 2026). Distilling Step-by-Step distinguishes rationale from output supervision, and AgentInstruct synthesizes teaching data through agentic flows (Hsieh et al., 2023; Mitra et al., 2024). Our small-model study uses final teaching artifacts for SFT and outcome-based GRPO (Shao et al., 2024); its learned target differs from supervising tool calls or full environment interactions.

Evaluation of open-ended artifacts. LLM judges scale response evaluation but can introduce systematic biases (Zheng et al., 2023). G-Eval and Prometheus use explicit criteria and structured scoring, while JudgeBench emphasizes objective error detection (Liu et al., 2023; Kim et al., 2024; Tan et al., 2025). We adapt QUEST's joint candidate/reference comparison (Xie et al., 2026) and retain both component scores. Following the diagnostic role of breakdowns in EduIllustrate (Bi et al., 2026), we examine agreement separately across educational Skill families and at the component and relative-score levels. This establishes where an evaluator agrees with reviewers, rather than treating a single aggregate as universal validation.

3 EduSkillBench

3.1 Task Definition and Construction

Given a teaching situation and request, an agent produces an instructional artifact: for example, a lesson plan, diagnostic question, support sequence, project brief, or practice schedule. Successful work requires interpreting the available evidence, choosing appropriate teaching operations, and delivering a correct, feasible product. Each task binds the situation and request to a designated Skill and a fixed set of applicable rubric dimensions. The Skill supplies reusable procedural support; it is not a reference answer.

The unified inventory contains 305 tasks across 14 Skills. It combines 263 Chinese teaching scenarios adapted from 30 documents in the archived `edu_scene_66` collection with 42 constructed English tasks, three per Skill. The source scenarios cover nine families; the constructed tasks extend coverage to all 14. Versioned revisions repair identified mismatches between questions, contexts, and criteria. Each constructed task specifies the requested artifact and constraints at one of three nominal difficulty levels. These levels organize construction, rather than constitute a calibrated proficiency scale. All 305 tasks receive equal weight in the primary result.

The packages are drawn from public educational Skill collections (Manning, 2026; YujxZJCN, 2026). They cover planning and alignment, diagnostic questioning, adaptive support, learner participation, and practice over time. Family sizes range from three to 157 tasks; lesson building accounts for 51.48% of the inventory. Figure 2 reports all family counts, and Appendix B preserves the full task and training distribution.

An example of procedure fit. One task asks how to help a fourth-grade pupil label rock specimens independently when labels are crooked or omit names and collection sites. Its adaptive-hint Skill connects diagnosis to graduated assistance and an independence check. The response must provide usable prompts and checks suited to this obstacle; reproducing an entire teaching template is unnecessary. The applicable rubric covers accuracy, scaffolding, guidance, age suitability, and safety. This task–procedure–criterion binding makes the intended support observable. It does not provide an answer key for every new example the agent may invent.

3.2 Paired Agent Execution

The native protocol uses an isolated Docker workspace for each task and OpenCode 1.18.11 with an educational single-turn agent instruction. NO SKILL exposes no educational package. WITH SKILL makes the designated task-matched package available through the native Skill interface. The agent decides whether to load it, inspect resources, and apply particular operations. This measures autonomous use of the available candidate; it does not test retrieval across all 14 Skills. Rubrics and evaluation references are withheld from generation.

The agent deadline is 600 seconds, with a 480-second provider timeout and a 10,000-token output cap per model call. Multiple calls may occur within a trajectory. Paired runs retain the requested artifact, available tool records, and execution status. There is one accepted outcome per model–task–condition, and completed answers are reused when only judging needs recovery. Exact model

settings and the retained GLM-5.3 exposure inventory are documented in Appendix A.

3.3 Reference-Conditioned Artifact Evaluation

DeepSeek V4.1 Flash jointly evaluates each candidate and the task's saved DeepSeek V4 Pro NO SKILL answer. The two texts appear as anonymous A/B responses, with hash-assigned order, the task, and its applicable rubric. The evaluator sees the retained delivery text; files absent from that input are not automatically inspected. Following the comparison mechanism of QUEST (Xie et al., 2026), both receive dimension scores $s_{x,i,d} \in [0, 10]$. We compute

$$J_{x,i} = \frac{1}{10D_i} \sum_{d=1}^{D_i} s_{x,i,d}, \tag{1}$$

$$R_i = \frac{J_{c,i}}{J_{c,i} + J_{r,i}}, \qquad S = \frac{100}{305} \sum_{i=1}^{305} R_i, \tag{2}$$

where D_i is the number of applicable dimensions and c, r denote candidate and reference. All tasks use this equal-weight rule, regardless of their construction origin. $R_i = 0.5$ means parity with the reference under that judgment, not 50% accuracy. Pro's reference-self condition receives 0.5; execution failures receive zero in the fixed 305-task denominator. A pair with both components zero is unresolved; none remain in the reported archive.

The judge checks correctness, consistency, feasibility, and task constraints; scores of at least 9 require response evidence. Substantive errors reduce relevant dimensions. Structural validators check coverage, evidence bindings, and score ranges. The reference is a comparison anchor, not a gold answer: generated scientific examples have no independent computational oracle. We therefore retain both J components and compare judgments with expert adjudication in Section 4.5. A stable ratio can conceal shared changes in candidate and reference scores.

4 Experimental Results

4.1 Experimental Setup

We evaluate GLM-5.3, GLM-5.3 Flash, DeepSeek V4.1 Flash, DeepSeek V4 Pro, and Doubao Seed 2.1 Pro on the same 305 tasks in both conditions. All 3,050 cells have a resolved score or an explicit execution-failure status. The native-access analysis covers the latter four systems; the archived GLM-5.3 row is retained in outcome comparisons, with its exposure provenance recorded in Appendix A. Qwen3-4B and its SFT/GRPO adapters form a separate learning comparison without external Skills.

The experiments ask three connected questions: how outcomes vary across Skills, how observed access relates to the resulting artifact, and whether Skill-informed demonstrations support performance without external resources. Expert-adjudicated judgments and cost measurements assess the reliability and practical requirements of these comparisons. All reported score changes are on the 0–100 reference-relative scale of Equation 2.

4.2 Overall Performance and Skill Breakdown

Procedural availability produces selective gains. Table 1 covers 3,050 model–task–condition cells: 2,725 accepted judgments, 305 reference-self comparisons, and 20 execution failures. GLM-5.3, GLM Flash, and Doubao improve by 0.45, 0.41, and 0.50 points with Skill availability, whereas DeepSeek Flash and Pro decline by 1.19 and 0.25 points. The small aggregate changes motivate examining which educational procedures help, whether agents access them, and what they ultimately deliver. They do not support a uniform benefit from adding procedural resources.

Different procedures reveal different strengths. Figure 2 groups the same accepted scores by designated Skill, without changing the judge or task membership. GLM-5.3, GLM Flash, DeepSeek Flash, Pro, and Doubao have positive differences on 10, 11, 7, 5, and 11 of 14 families. For example, GLM Flash gains 0.95 points on Socratic questioning but loses 0.81 on retrieval practice. DeepSeek Flash gains 1.01 on self-efficacy support and 0.98 on Socratic questioning, yet loses 2.49 on lesson building. These contrasts locate procedure-specific opportunities that the overall mean conceals.

Coverage and completion explain part of the variation. Lesson building contributes 157/305 tasks (51.48%), while five families contain three tasks each. Its decline contributes -1.28 points to DeepSeek Flash's overall -1.19 change. Equal weighting of the 14 families instead gives changes of $+0.47$, $+0.34$, -0.09, -0.13, and $+1.87$

Model	No Skill	With Skill	Δ (points)	Failures (No/With)
GLM-5.3	50.73	51.19	+0.45	2/1
GLM-5.3 Flash	50.15	50.56	+0.41	1/0
DeepSeek V4.1 Flash	50.70	49.51	-1.19	1/5
DeepSeek V4 Pro	50.00	49.75	-0.25	0/1
Doubao Seed 2.1 Pro	50.63	51.13	+0.50	5/4

Table 1: Reference-relative scores on the unified 305-task inventory. All tasks remain in each denominator; execution failures contribute zero. Pro No Skill supplies the fixed references and is assigned the self-comparison value 50.00. GLM-5.3 is the retained augmentation result described in Appendix A. Differences use unrounded values.

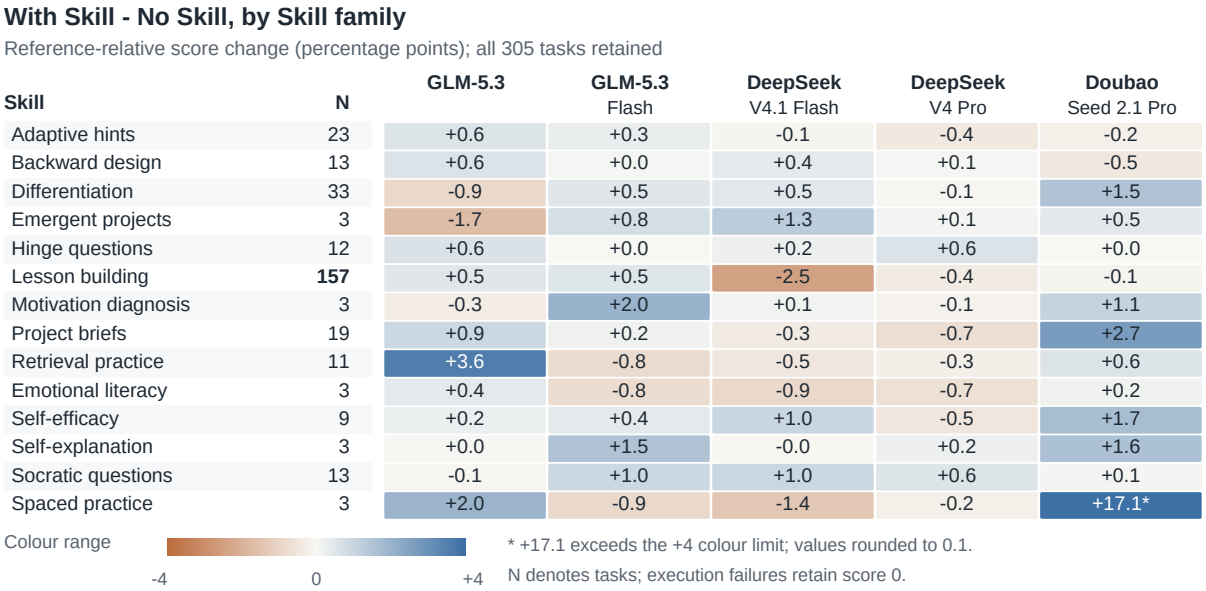


Figure 2: With–No score differences for all five models across the 14 Skills. Each cell retains all tasks in its family, including failures; n gives the family size. Color saturates at ± 4 points; labels show rounded differences. Complete means and failure counts appear in Appendix B.

points in the model order above; this diagnostic describes coverage sensitivity, while the task-weighted mean remains primary.

Completion also matters. Doubao's spaced-practice gain is $+17.06$ on three tasks, including one failed No Skill execution; the two pairs completed in both conditions differ by only $+0.59$. Across all common-valid pairs, the four native-access models change by $+0.25$, -0.54, -0.09, and $+0.36$ points, respectively. DeepSeek Flash's decline therefore persists after conditioning on completion. Appendix B supplies family denominators and failure diagnostics. These single-run differences describe the retained executions; they do not establish significance across model or judge resampling.

4.3 From Skill Access to Usable Teaching Artifacts

The native interface exposes the task's designated educational package; the agent decides whether to load it. We count a completed call naming that package, requiring returned content for locally available raw traces and using completed-call summaries where those are the retained evidence. Missing captures remain unknown. GLM-5.3 is excluded from this access analysis because its retained answers span two exposure modes (Appendix A).

Availability does not guarantee access. Figure 3 records 531 loads over 1,220 With tasks, ranging from Pro's 17.70% to DeepSeek Flash's 71.48%. Every recorded educational load names the designated package. With one task-scoped educational candidate, this measures the decision to access relevant guidance; it does not test retrieval among all 14 Skills. Loaded common-valid groups improve for GLM Flash, Pro, and Doubao, but decline for DeepSeek Flash. Since agents choose whether to load and the resulting groups differ in task composition, these associations do

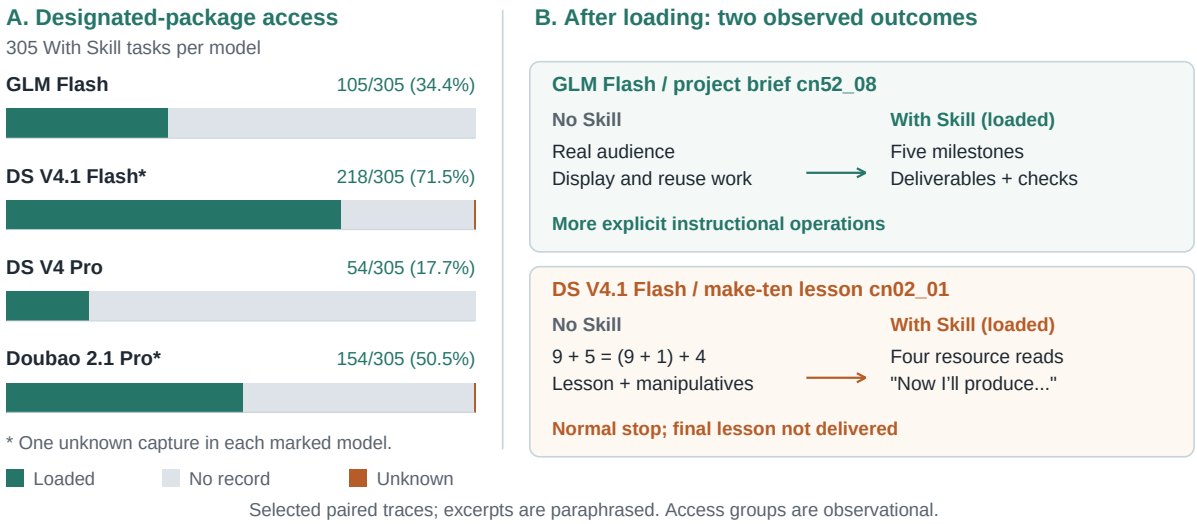


Figure 3: From package availability to teaching artifacts. Left: observed access among 305 With tasks per model; unknown captures are separate from no recorded access. Right: paraphrased paired artifacts show explicit project milestones after loading, but a lesson that ends as a promise rather than a deliverable. Cases illustrate mechanisms, not prevalence or a causal effect of loading. Full subgroup results appear in Appendix B.

not identify a causal loading effect.

A useful procedure makes classroom operations explicit. In GLM Flash's project-brief task `cn52_08`, the teacher has no budget or laboratory and wants student projects with a real audience. Both answers identify meaningful uses for the work. After loading the package, the With answer additionally organizes five milestones, stage-specific deliverables, explicit instruction, and completion checkpoints. This is observable procedural uptake beyond mentioning the Skill. Judged candidate quality J rises from 0.856 to 0.912, although the reference component also falls from 0.918 to 0.806; the resulting $+4.83$ relative-score change therefore cannot be assigned entirely to the candidate's improvement.

Successful access can still end without a deliverable. For DeepSeek Flash's lesson-building task `cn02_01`, No Skill provides a concrete make-ten lesson with manipulatives and fallback materials. With Skill loads the package and four resources, but terminates with a promise to produce materials instead of the lesson itself. The answer has a normal stop and a zero candidate component. More broadly, DeepSeek Flash has 60 failed supporting-file reads over 49 tasks despite successful main package loads. These records distinguish access, resource execution, and final delivery. Our 32 exported paired cases connect those stages to actual teaching products; they were selected diagnostically and do not estimate failure prevalence (Appendix B).

4.4 Learning to Produce Artifacts Without External Skills

We further ask whether Skill-informed demonstrations can improve a small model when the procedural documents are absent at inference. Inspired by SkillGym's distinction between external assistance and parameter transfer (Ge et al., 2026), we compare Qwen3-4B-Instruct-2507, LoRA-SFT, and subsequent outcome-based GRPO on the same 305 tasks under No Skill.

Teach decisions through complete artifacts. We decompose the 14 Skills into operations and evidence that should change their application: repeated failed hints, for example, should trigger a different response from an initial request for help. EduBench question seeds are reconstructed around complete-product creation, changed learner evidence, resource boundaries, and artifact repair. Screening and revision retain 246 demonstrations: 162 generated reference artifacts, 52 revised artifacts, and 32 final answers from environment executions. Each student input contains the task; the supervised target is its complete teaching artifact. Skill documents and rubrics are omitted from student inputs. This supervision targets educational decisions expressed in products, rather than native tool-call trajectories.

Stage	No Skill	Δ_{Base}	Δ_{prev}
Base	41.56	–	–
SFT246	42.69	+1.13	+1.13
SFT246 → GRPO123	43.04	+1.48	+0.35

Table 2: Reported 305-task No Skill aggregates for Qwen3-4B and its two adapters. The latest per-task recovery archive remains to be reconciled with this supplied aggregate version (Appendix A).

Score	MAE	Bias	ρ
Candidate J_c	8.61	-6.13	0.251
Reference J_r	7.43	-4.04	0.086
Relative R	1.99	-0.58	0.165

Table 3: 249-pair agreement. MAE and human-minus-Judge bias are in percentage points; ρ is Spearman correlation.

SFT applies completion-only loss for one epoch, preserving all complete targets within 8,192 tokens. GRPO starts from that adapter, samples eight answers per training question, and runs 123 optimization steps. The training judge assigns the equally weighted PASS fraction under the task rubric; empty or truncated answers receive zero. This outcome reward differs from the final reference-relative evaluation. Full recipes and provenance are in Appendix A.

The reported sequence is 41.56→42.69→43.04: $+1.13$ points after SFT and another $+0.35$ after GRPO. This is a modest descriptive improvement, not a demonstrated routing capability or a significant effect. Training uses non-identical benchmark-derived variants, so the comparison concerns related educational families rather than unseen-family generalization. Updated adapter evaluations also recover historical execution failures; item-level reconciliation is needed to separate recovery and answer-quality contributions before finalizing the learning claim.

4.5 Human–Judge Agreement

Design. Two expert reviewers jointly adjudicated 28 tasks, two per Skill, covering five models in both conditions. Excluding 28 reference-self and three execution-failure controls leaves 249 candidate–reference pairs. Reviewers use the same rubric and 0–10 scale, scoring each reference once. Following Bi et al. (2026), we examine agreement by educational dimension; here the breakdown follows 14 Skill families (Appendix C).

Score proximity and ranking differ. Table 3 shows a 1.99-point R MAE, compared with 8.61 and 7.43 for candidate and reference quality. Candidate J averages 92.23 under the Judge and 86.10 after adjudication. Ratio compression and cancellation of shared shifts can reduce this discrepancy without validating absolute correctness or improving pointwise judging. The R correlation is 0.165; exact candidate/tie/reference agreement is 133/249 (53.41%).

Disagreement is skill-specific. Figure 4 localizes the largest R errors to hinge questions (4.26 points) and retrieval practice (3.70), versus adaptive hints (1.10) and emergent projects (1.17). Faulty answer keys undermine worked solutions and student diagnoses despite plausible teaching structure. Backward design shows stronger relative ordering ($\rho = 0.756$). These differences identify teaching operations that need closer inspection. Each family contributes only two reviewed tasks, and the experts produce one consensus label set rather than independent ratings. The analysis therefore diagnoses evaluator behavior without establishing universal reliability or resolving small leaderboard differences.

4.6 The Cost of Procedural Support

Following stage-level accounting in EduIllustrate (Bi et al., 2026), we sum model-call usage within each task before comparing costs. On three fixed tasks with retained native traces, GLM Flash's mean calls increase from 1.00 to 2.00 with Skill access, input tokens from 5,758 to 14,998, and estimated cost from CNY 0.01283 to 0.02062: 2.60× the input and 1.61× the cost. DeepSeek Flash's With executions average 4.33 calls and 41,776 input tokens. These examples expose repeated-context overhead behind the access decisions, without estimating the full-inventory mean.

A separate direct-API pilot makes 12 requests, four models on the same three tasks, at a 4,096-token cap. Its public-tariff estimate totals CNY 0.6430; seven responses stop naturally and five reach the cap. Accordingly, this measures bounded request cost, not cost per correct teaching artifact. The pilot omits the native harness, and its provider configuration is scoped separately. Appendix D gives model versions, per-model results, cached-token accounting, and dated rates (Volcano Engine, 2026).

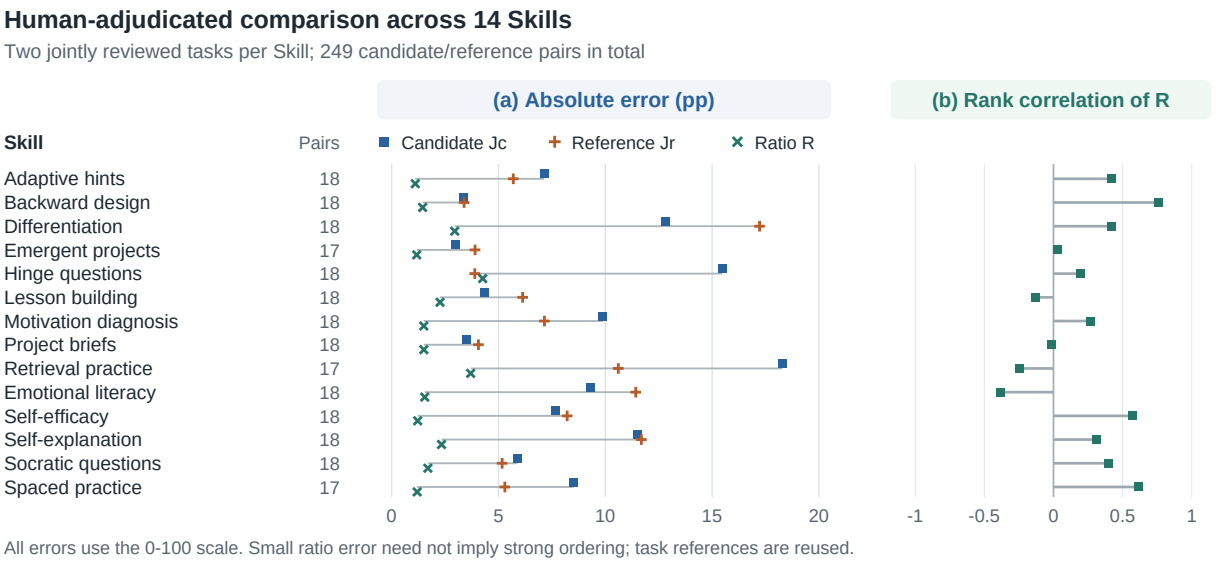


Figure 4: Human–Judge comparison across 14 Skills: two reviewed tasks per Skill and 249 substantive pairs. Left: candidate J_c, reference J_r, and relative-score R MAE (percentage points). Right: R Spearman correlation. References are reused within tasks; small ratio errors can coexist with weak ordering agreement.

5 Conclusion

EDUSKILLBENCH evaluates the connection between educational procedures, agent behavior, and delivered teaching artifacts. Its 305 tasks and 14 Skill families support paired outcome comparisons, observable package access, and family-level diagnosis. The results show useful procedural detail in selected cases alongside incomplete delivery and resource failures after access. Small-model artifact learning provides a complementary route to studying performance without external documents. Expert-adjudicated breakdowns clarify where reference-conditioned judgments agree and where their components differ. Together, these analyses turn a coarse question—whether Skills help—into testable questions about access, appropriate operations, and usable products.

Limitations

The 305-task inventory is unbalanced: lesson building contributes over half the tasks, while five families contain three each. Task origin, language, and rubric style are correlated. Access is self-selected, so loaded versus non-loaded groups do not estimate a causal effect of reading a Skill. Full raw traces are locally available for two native models; the other two have retained tool summaries. Provider implementations and a single retained outcome per cell limit conclusions about repeated-run variability.

Our fixed reference can contain errors, and joint scoring can depend on its paired candidate. The 28-task review covers all Skills but is not a proportionate sample of the 305-task inventory. Two experts jointly approved one label set, providing no independent inter-rater reliability estimate. Benchmark-derived training variants test related-family transfer rather than unseen-family generalization; the latest small-model recovery aggregates still require item-level reconciliation. Final-artifact supervision does not directly teach native package calls. None of these scores measures student learning outcomes or establishes deployment safety.

Ethical Considerations

The study uses public procedural resources and de-identified teaching situations. Generated artifacts can contain misleading examples or inappropriate advice and require substantive review before classroom use. Release materials retain source attribution and evaluation provenance while omitting credentials and reviewer affiliations. The expert review procedure and prior assistance in preparing its labels are documented in Appendix C.

References

Agent Skills. 2026. Agent skills specification. Accessed October 5, 2026.

Shuzhen Bi, Mingzi Zhang, Zhuoxuan Li, Xiaolong

Wang, Keqian Li, and Aimin Zhou. 2026. EduIllustrate: Towards scalable automated generation of multimodal educational content. *arXiv preprint arXiv:2604.05005*.

Zhilong Ge, Yuting Shao, Yutao Yang, Yuxuan Cai, Jie Zhou, Kai Chen, Bo Zhang, Qin Chen, and Liang He. 2026. SkillGym: Internalizing large-scale human skills into LLMs for real-world problem solving. *arXiv preprint arXiv:2609.27717v1*.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017.

Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. 2024. Prometheus: Inducing fine-grained evaluation capability in language models. In *International Conference on Learning Representations*.

LearnLM Team. 2024. LearnLM: Improving gemini for learning. *arXiv preprint arXiv:2412.16429*.

Xiangyi Li, Yimin Liu, Wenbo Chen, et al. 2026. Skillsbench: Benchmarking how well agent skills work across diverse tasks. *arXiv preprint arXiv:2602.12670v4*.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522.

Zhengxi Lu, Zhiyuan Yao, Jinyang Wu, Chengcheng Han, Qi Gu, Xunliang Cai, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. 2026. SKILL0: In-context agentic reinforcement learning for skill internalization. *arXiv preprint arXiv:2604.02268v2*.

Jakub Macina, Nico Daheim, Ido Hakimi, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2025. MathTutorBench: A benchmark for measuring open-ended pedagogical capabilities of LLM tutors. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 204–221.

Gareth Manning. 2026. Education agent skills library. Third-party Skill source; archived attribution in the benchmark.

Kaushal Kumar Maurya, Kv Aditya Srivatsa, Kseniia Petukhova, and Ekaterina Kochmar. 2025. Unifying AI tutor evaluation: An evaluation taxonomy for pedagogical ability assessment of LLM-powered AI tutors. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1234–1251.

Arindam Mitra, Luciano Del Corro, Guoqing Zheng, Shweti Mahajan, Dany Rouhana, Andres Codas, Yadong Lu, Wei-ge Chen, Olga Vrousgos, Corby Rosset, Fillipe Silva, Hamed Khanpour, Yash Lara, and Ahmed Awadallah. 2024. AgentInstruct: Toward generative teaching with agentic flows. *arXiv preprint arXiv:2407.03502v1*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Lee S. Shulman. 1986. Those who understand: Knowledge growth in teaching. *Educational Researcher*, 15(2):4–14.

Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y. Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. 2025. JudgeBench: A benchmark for evaluating LLM-based judges. In *International Conference on Learning Representations*.

Volcano Engine. 2026. Model pricing: Regular online inference. Public tariffs observed 12 October 2026; currency CNY.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024a. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*.

Rose Wang, Qingyang Zhang, Carly Robinson, Susanna Loeb, and Dorottya Demszky. 2024b. Bridging the novice-expert gap via models of decision-making: A case study on remediating math mistakes. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, pages 2174–2199.

Rose E. Wang, Ana T. Ribeiro, Carly D. Robinson, Susanna Loeb, and Dorottya Demszky. 2025. Tutor CoPilot: A human-AI approach for scaling real-time expertise. Technical report, Annenberg Institute, Brown University. November 2025 revision.

David Wood, Jerome S. Bruner, and Gail Ross. 1976. The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17(2):89–100.

Jian Xie, Tianhe Lin, Zilu Wang, Yuting Ning, Yuekun Yao, Tianci Xue, Zhehao Zhang, Zhongyang Li, Kai Zhang, Yufan Wu, Shijie Chen, Boyu Gou, Mingzhe Han, Yifei Wang, Vint Lee, Xinpeng Wei, Xiangjun

Wang, Yu Su, and Huan Sun. 2026. QUEST: Training frontier deep research agents with fully synthetic tasks. *arXiv preprint arXiv:2605.24218*.

Bin Xu, Yu Bai, Huashan Sun, Yiguan Lin, Siming Liu, Xinyue Liang, Yaolin Li, Zhuangzhi Dong, Jingren Zhang, Yufan Deng, Xinyu Zou, Yang Gao, and Heyan Huang. 2025. Edubench: A comprehensive benchmarking dataset for evaluating large language models in diverse educational scenarios. *arXiv preprint arXiv:2505.16160*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*.

YujxZJCN. 2026. Teaching skills. Third-party Skill source; archived attribution in the benchmark.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*.

A Reproducibility and Version Boundaries

A.1 Generation Configuration

All current result rows use the frozen 305-task objects in the same order and the reference-relative protocol of Section 3.3. The four native model groups share the image with SHA-256 prefix `1f72e58f6d9b5176` (full digest in the release manifest), OpenCode 1.18.11, and the education-single-turn agent prompt. Each condition retains a per-task workspace and logs. Native With exposes the designated candidate package, not all 14 packages. No rubric or evaluation reference is given to generation. The package mount and any available supporting files are part of the release manifest.

The provider aliases are `glm-5-3-flash-260828`, `deepseek-v4.1-flash`, `deepseek-v4-pro-ga-260813`, and `doubao-seed-2-1-pro-260915`. GLM Flash, Pro, and Doubao use the Ark `/api/v3` protocol; DeepSeek Flash uses a third-party OpenAI-compatible `/v1` endpoint. GLM Flash enables low-effort thinking, whereas the other three disable it. These recorded switches are not claims about hidden provider computation. Environment concurrency is 8 for GLM Flash and DeepSeek Flash and 32 for Pro and Doubao; the API concurrency ceiling is 64. Scheduling concurrency is not repeated sampling or a per-task roll-out count.

The retained GLM-5.3 row has 262 With answers from full-text Skill exposure with native resources and 43 from native candidate-package exposure. It is a historical composite result under the current task inventory and score metric, not a fifth matched native-exposure intervention. It is retained for result coverage but excluded from native-access statistics. The task-level exposure manifest is necessary to reproduce this row; an aggregate score cannot identify the effect of a single exposure method.

A.2 Judge Inputs and Aggregation

The accepted judge uses `deepseek-v4.1-flash`, thinking disabled, 6,000 maximum output tokens, and a final recovery timeout of 600 seconds. The fixed reference answer text is the archived Pro No Skill answer for each task, not a human gold answer. Candidate and reference are anonymized jointly; the rubric scope, segmentation, A/B ordering, and hashes are retained. High-scoring evidence citations refer to the appropriate response's own segments. Score validation checks structure and bindings, not the truth of the quoted claims. Each accepted pair retains candidate and reference dimension scores, evidence, and the derived ratio.

The primary calculation is identical for all 305 tasks: equally average applicable 0–10 dimensions within each response, compute its per-task candidate/reference ratio, and average those ratios over the full inventory. An execution failure contributes zero; a reference-self cell contributes 0.5. Missing or invalid verdicts must remain unresolved, not be assigned convenient defaults. None remain in the five-model archive. Transport retries and rejected schema responses may precede an accepted verdict, so one retained verdict does not imply exactly one historical HTTP request.

A.3 Historical Rubric Metadata

The source inventory contains 60 tasks with three native grades, 100 with four, 20 with two grades plus a below-minimum fallback, and 83 with satisfaction labels where native grades are absent. The remaining 42 constructed tasks have weighted binary items. This provenance is retained, but the current 305-task metric does not count top grades on one subset and all-pass outcomes on another.

The current human-adjudicated comparison in Appendix C uses the same continuous dimensions and paired answers as the deployed Ref judge, not those earlier aggregation formulas. A pointwise score and a normalized relative score are on different scales; their numerical difference does not measure a quality change or a correction of inflation.

A.4 Small-Model Provenance

Training seeds use the EduBench snapshot at commit `f19004a5c68c27cd9d35 4401654345072fb712c6`. This is a training-seed source, not a claim that every evaluation task is an official EduBench item. The 246-example release contains 80 Skill–seed combinations from 70 upstream seeds. Variant counts are 62 complete-product tasks, 65 changed-evidence tasks, 61 resource-constraint tasks, and 58 artifact-repair tasks. The recorded `environment_trace_verified` flag is false for all 246 examples; the retained targets are complete final answers, not verified full trajectories.

The SFT base is Qwen3-4B-Instruct-2507. Its one-epoch LoRA run uses bf16 and completion-only labels, rank 32, alpha 64, dropout 0.05, learning rate 10^{-5}, and cosine scheduling with 5% warmup. Adapters update attention $q/k/v/o$ and MLP gate/up/down projections. Micro-batch 8 with accumulation 2 gives an effective batch of 16; training completes 16 optimizer steps. The longest retained training sequence has 7,723 tokens, within the 8,192-token limit; the mean assistant target has 1,091.81 tokens. The run takes approximately 239 seconds. GRPO completes 123 optimizer steps at learning rate 2×10^{-6}, using eight samples per question and generation batch 16, with a 4,096-token output limit, temperature 0.7, top-p 0.9, cosine scheduling, 5% warmup, and a DAPO-style loss. Its frozen KL policy is the SFT adapter with coefficient 0.04; empty and truncated outputs receive zero reward. Its run takes approximately 3,536 seconds. The training outcome judge is `deepseek-v4-flash-ga-260731`; the evaluation judge is V4.1 Flash. These are different model versions and reward scales, as specified in Section 4.4.

The current review draft's small-model table is bound to the supplied recovery summary, while the latest item-level recovery archive is pending local reconciliation. That table should not be finalized for submission until the answer hashes, accepted verdicts, failure states, and paired differences are checked. The five API-system tables and native-access tables have already been recomputed from their retained item-level materials.

Skill family	Test tasks	SFT examples
Adaptive hints	23	20
Backward design	13	22
Differentiation	33	21
Emergent projects	3	19
Hinge questions	12	18
Lesson building	157	15
Motivation diagnosis	3	11
Project briefs	19	21
Retrieval practice	11	18
Emotional literacy	3	18
Self-efficacy	9	20
Self-explanation	3	23
Socratic questions	13	16
Spaced practice	3	4
Total	**305**	**246**

Table 4: Unified benchmark and training inventory. Family coverage is complete but sizes and task distributions are not balanced or proportional. Counts do not establish coverage of every principal operation within a Skill.

A.5 Release Contents

The retained experiment archive includes task/source/Skill IDs, exposure modes, model request aliases, generation and judging settings, accepted responses, failure states, rubric scopes, joint judgments, access evidence tiers, and offline aggregation scripts. The manuscript bundle contains compact frozen tables, case exports, and provenance manifests. These materials separate original pointwise experiments from current reference-relative experiments. Credentials, personal server routes, and unrelated projects are excluded. Availability of a record is not silently promoted to evidence of successful file delivery or scientific correctness.

B Coverage and Access Diagnostics

The five families with three benchmark tasks only are emergent projects, motivation diagnosis, emotional literacy, self-explanation, and spaced practice. Lesson building contributes 154 source scenarios and three constructed tasks. The underlying task objects and order are unchanged between the frozen export and the present scoring archive, despite JSON formatting differences in their file hashes.

Skill family	GLM Flash	DS V4.1 Flash	DS V4 Pro	Doubao 2.1 Pro
Adaptive hints	20/23	9/23	3/23	13/23
Backward design	5/13	9/13	1/13	6/13
Differentiation	4/33	15/33	3/33	5/33
Emergent projects	3/3	3/3	3/3	3/3
Hinge questions	4/12	4/12	3/12	10/12
Lesson building	29/157	130/157	18/157	66/157
Motivation diagnosis	3/3	3/3	3/3	3/3
Project briefs	7/19	7/19	4/19	12/19
Retrieval practice	4/11	9/11	4/11	7/11
Emotional literacy	3/3	3/3	3/3	3/3
Self-efficacy	5/9	7/9	2/9	9/9
Self-explanation	2/3	3/3	3/3	3/3
Socratic questions	13/13	13/13	3/13	11/13
Spaced practice	3/3	3/3	1/3	3/3

Table 5: Completed expected-package loads / available tasks. Unknown captures are retained in the available-task denominator. Absence of a competing educational package means these counts measure access, not full-library retrieval accuracy.

B.1 Complete Model–Skill Score Breakdown

Tables 7 and 8 retain the No/With means behind every difference in Table 6. Each mean uses all n_s tasks in that family, including zero-valued execution failures. The same fixed task membership is used for all five models. Table 9 reports the corresponding execution failures; a completed answer that receives a low rubric score is not an execution failure. Per-task answer bindings, both judgment components, common-valid denominators, paired-direction counts, and weighted contributions to the 305-task mean are retained in the offline analysis release.

B.2 Evidence Levels and Case Selection

Raw-trace access requires a completed designated Skill call with returned content in the root answer's execution tree. Summary-level access requires a retained completed-call record with the designated name but permits less inspection of subsequent resource actions. The two unknown captures concern DeepSeek Flash's `lesson-builder__cn01_05` and Doubao's `lesson-builder__03`. They are not counted as proven non-access. One No Skill GLM Flash trace loads a built-in OpenCode customization package; this is not an exposed educational Skill and is not counted as such.

The process-analysis bundle includes 32 case exports with saved No/With answers, judgment components, and available tool evidence. Cases illustrate positive, negative, tied, and incomplete outcomes; they do not estimate a population error rate. The native records distinguish main-package loading from supporting-file reads. In DeepSeek Flash, the 60 read errors comprise 41 pedagogy, 16 checkpoint, two quality-gate, and one course-passport reference failures. Their frequency is specific to this mounted resource inventory.

B.3 Candidate and Reference Component Changes

The same reference text may receive different component scores in separate candidate pairs. Positive ratio changes can therefore accompany decreases in the judged reference component. Retaining these quantities prevents normalization from being mistaken for an independent measurement of absolute improvement. Neither component analysis nor anonymous presentation proves immunity to position, length, self-family, or reference-quality biases.

Skill family	n_s	GLM-5.3	GLM Flash	DS Flash	DS Pro	Doubao
Adaptive hints	23	+0.584	+0.297	-0.088	-0.424	-0.236
Backward design	13	+0.588	+0.044	+0.410	+0.051	-0.481
Differentiation	33	-0.886	+0.511	+0.477	-0.065	+1.542
Emergent projects	3	-1.672	+0.759	+1.292	+0.099	+0.454
Hinge questions	12	+0.626	+0.009	+0.215	+0.588	+0.000
Lesson building	157	+0.501	+0.519	-2.491	-0.353	-0.138
Motivation diagnosis	3	-0.333	+2.008	+0.143	-0.143	+1.086
Project briefs	19	+0.926	+0.159	-0.348	-0.729	+2.714
Retrieval practice	11	+3.607	-0.808	-0.490	-0.278	+0.575
Emotional literacy	3	+0.419	-0.783	-0.909	-0.681	+0.157
Self-efficacy	9	+0.249	+0.427	+1.008	-0.485	+1.732
Self-explanation	3	+0.029	+1.537	-0.048	+0.212	+1.594
Socratic questions	13	-0.066	+0.955	+0.982	+0.561	+0.069
Spaced practice	3	+1.993	-0.879	-1.374	-0.227	+17.058

Table 6: With–No score differences across all 14 Skills. Every family retains its full task denominator, including execution failures. These are the numerical values behind Figure 2; additional decimals expose rounding, not statistical certainty.

		GLM-5.3			GLM Flash			DS V4.1 Flash		
Skill family	n_s	No	With	Δ	No	With	Δ	No	With	Δ
Adaptive hints	23	50.507	51.091	+0.584	49.936	50.232	+0.297	50.829	50.740	-0.088
Backward design	13	51.051	51.639	+0.588	50.482	50.525	+0.044	51.868	52.278	+0.410
Differentiation	33	51.558	50.672	-0.886	49.904	50.415	+0.511	50.423	50.900	+0.477
Emergent projects	3	52.344	50.672	-1.672	50.668	51.427	+0.759	49.676	50.967	+1.292
Hinge questions	12	50.757	51.383	+0.626	49.806	49.814	+0.009	50.955	51.170	+0.215
Lesson building	157	50.821	51.322	+0.501	50.201	50.719	+0.519	50.553	48.062	-2.491
Motivation diagnosis	3	51.019	50.686	-0.333	50.154	52.162	+2.008	51.135	51.278	+0.143
Project briefs	19	50.268	51.194	+0.926	49.753	49.912	+0.159	50.489	50.141	-0.348
Retrieval practice	11	46.856	50.463	+3.607	51.166	50.358	-0.808	51.865	51.375	-0.490
Emotional literacy	3	50.223	50.642	+0.419	50.660	49.877	-0.783	51.431	50.521	-0.909
Self-efficacy	9	50.925	51.174	+0.249	49.961	50.388	+0.427	50.009	51.017	+1.008
Self-explanation	3	51.116	51.145	+0.029	48.990	50.527	+1.537	50.698	50.651	-0.048
Socratic questions	13	51.353	51.287	-0.066	50.444	51.399	+0.955	51.040	52.022	+0.982
Spaced practice	3	49.631	51.624	+1.993	50.260	49.381	-0.879	51.871	50.497	-1.374

Table 7: Complete fourteen-Skill breakdown for the three result inventories shown: reference-relative No/With means on the 0–100 scale and differences in score points. Deltas are calculated from unrounded means. Scores are not scientific-accuracy percentages.

		DS V4 Pro			Doubao Seed 2.1 Pro		
Skill family	n_s	No	With	Δ	No	With	Δ
Adaptive hints	23	50.000	49.576	-0.424	51.222	50.985	-0.236
Backward design	13	50.000	50.051	+0.051	51.753	51.272	-0.481
Differentiation	33	50.000	49.935	-0.065	50.299	51.841	+1.542
Emergent projects	3	50.000	50.099	+0.099	33.683	34.137	+0.454
Hinge questions	12	50.000	50.588	+0.588	51.564	51.564	+0.000
Lesson building	157	50.000	49.647	-0.353	51.534	51.395	-0.138
Motivation diagnosis	3	50.000	49.857	-0.143	49.843	50.929	+1.086
Project briefs	19	50.000	49.271	-0.729	45.963	48.678	+2.714
Retrieval practice	11	50.000	49.722	-0.278	51.492	52.067	+0.575
Emotional literacy	3	50.000	49.319	-0.681	49.902	50.059	+0.157
Self-efficacy	9	50.000	49.515	-0.485	51.325	53.058	+1.732
Self-explanation	3	50.000	50.212	+0.212	50.036	51.630	+1.594
Socratic questions	13	50.000	50.561	+0.561	51.238	51.307	+0.069
Spaced practice	3	50.000	49.773	-0.227	34.431	51.489	+17.058

Table 8: Complete fourteen-Skill breakdown for Pro and Doubao, with the same denominators and scoring definition as Table 7. Pro's No value is 50 in every family because it is the reference-self condition; it is not an independent judgment of Pro answer quality.

Skill family	n_s	GLM-5.3	GLM Flash	DS Flash	DS Pro	Doubao
Adaptive hints	23	0/0	0/0	0/0	0/0	0/0
Backward design	13	0/0	0/0	0/0	0/0	0/0
Differentiation	33	0/0	0/0	0/0	0/0	1/0
Emergent projects	3	0/0	0/0	0/0	0/0	1/1
Hinge questions	12	0/0	0/0	0/0	0/0	0/0
Lesson building	157	1/1	1/0	1/5	0/1	0/2
Motivation diagnosis	3	0/0	0/0	0/0	0/0	0/0
Project briefs	19	0/0	0/0	0/0	0/0	2/1
Retrieval practice	11	1/0	0/0	0/0	0/0	0/0
Emotional literacy	3	0/0	0/0	0/0	0/0	0/0
Self-efficacy	9	0/0	0/0	0/0	0/0	0/0
Self-explanation	3	0/0	0/0	0/0	0/0	0/0
Socratic questions	13	0/0	0/0	0/0	0/0	0/0
Spaced practice	3	0/0	0/0	0/0	0/0	1/0

Table 9: Execution failures in No/With order for every model–Skill group. These counts expose completion-driven changes without removing failures from the reported scores. A zero count does not mean the answers are correct, or that the designated package was used.

Model	Loaded	Rate (%)	Δ_L	Δ_N
GLM Flash	105	34.43	+0.584	+0.074
DS V4.1 Flash	218	71.48	-0.549	-0.507
DS V4 Pro	54	17.70	+0.098	-0.124
Doubao 2.1 Pro	154	50.49	+0.656	+0.065

Table 10: Designated-package loads among 305 With tasks. Δ_L and Δ_N are With–No changes on common-valid pairs with a load and with no recorded access. DeepSeek Flash and Doubao each have one unknown capture, retained in rate denominators.

	Loaded		No recorded access	
Model	N	Δ	N	Δ
GLM Flash	105	+0.584	199	+0.074
DS V4.1 Flash	213	-0.549	86	-0.507
DS V4 Pro	53	+0.098	251	-0.124
Doubao 2.1 Pro	147	+0.656	149	+0.065

Table 11: Common-valid subgroup denominators and With–No score differences for Table 10. Groups are selected by the agent's recorded access behavior; their differences do not identify a causal loading effect.

Model	Access group	N	ΔJ_{cand}	ΔJ_{ref}	ΔR
GLM Flash	Loaded	105	+0.801	-1.287	+0.584
GLM Flash	No record	199	+0.329	+0.009	+0.074
DS V4.1 Flash	Loaded	213	-1.878	-0.859	-0.549
DS V4.1 Flash	No record	86	-1.057	-0.225	-0.507
DS V4 Pro	Loaded	53	–	–	+0.098
DS V4 Pro	No record	251	–	–	-0.124
Doubao 2.1 Pro	Loaded	147	+0.833	-1.313	+0.656
Doubao 2.1 Pro	No record	149	+0.577	+0.248	+0.065

Table 12: With–No changes on common valid pairs grouped by With access. All three deltas are multiplied by 100; ΔR averages per-task ratios rather than applying the ratio to the two mean components. Pro's reference-self No condition has no judged component scores, shown as unavailable rather than imputed.

C Complete Human Adjudication and Skill Breakdown

C.1 Coverage and Review Procedure

The review binds saved answer text and accepted scores from the current 305-task Ref experiment. A deterministic task-ID hash selects two tasks per Skill, requiring at least eight usable model–condition cells out of ten. This yields 18 source-scenario tasks and ten constructed tasks. The allocation ensures Skill coverage rather than representing the benchmark's uneven frequency distribution. The five model groups and both access conditions match the main API comparison; GLM-5.3 retains its recorded historical execution provenance.

All 280 sampled cells are accounted for: 249 substantive pairs, 28 reference-self controls, and three execution-failure controls. Every substantive pair has a complete set of scores, textual evidence, counterevidence where identified, and reasons. Controls retain their predefined values and do not enter agreement statistics. No answers were regenerated and no additional Judge requests were made for this analysis.

The initial scores and evidence were AI-proposed. Two expert reviewers subsequently jointly reviewed and approved all 249 candidate label sets and 28 reference label sets, as confirmed by the study organizer. Affiliations are anonymized. The resulting human-adjudicated labels are one shared set; there are no separate independent reviewer scores. We therefore do not estimate inter-rater reliability or claim an independent multi-rater design.

Review inputs preserve the task, applicable rubric, original anonymous A/B order, and E-numbered delivery text, without displaying the saved numerical Judge scores. Proposal preparation had prior exposure to aggregates and some pilot cases; the joint human review is not documented as fully blinded. Each reference is scored once and its labels are reused across candidates. The data contain 1,345 candidate dimension scores and 152 unique reference dimension scores (1,497 labels). The expanded comparison has 2,690 rows, including reused references, not 2,690 independent human judgments.

C.2 Comparisons and Denominators

Both sides use the same applicable dimensions and continuous 0–10 scale. For D dimensions,

$$J_c = \frac{1}{10D}\sum_d s_{c,d}, \qquad J_r = \frac{1}{10D}\sum_d s_{r,d},$$

$$R = \frac{J_c}{J_c + J_r}.$$

Ratios are calculated per pair before averaging. Parity at 0.5 is not 50% accuracy; no pair has an undefined zero denominator. MAE and signed bias compare paired values, with bias defined as adjudicated minus Judge. Spearman correlation uses average ranks for ties and returns undefined for constant inputs. Statistics are descriptive: repeated model outputs and reference scores cluster within 28 tasks.

On 1,345 candidate dimensions, MAE is 0.992 on the 0–10 scale and Spearman correlation is 0.270. Scores of at least nine occur on 1,163 dimensions under the Judge and 780 after adjudication; 234 adjudicated dimensions still receive ten. Thus the review identifies localized disagreements without imposing a ceiling on valid high scores. Table 13 reports every Skill rather than selecting only favorable correlations.

C.3 Reference Effects and Error Interpretation

The same saved reference receives different automated scores across candidate contexts: its within-task J_r range averages 11.13 points and reaches 25.00. Comparing each task's mean automated reference score against its single adjudicated score gives a 6.88-point MAE over 28 unique references. The absence of within-task variation in adjudicated references follows from label reuse, not repeated-review stability.

Exact three-way preference agreement is 53.41%. Treating scores within one or two points of parity as ties changes agreement to 46.18% or 62.65%, respectively. These thresholds are sensitivity checks, not tuned choices for the headline result. Component differences and weak pooled ranking correlation explain why a small ratio MAE cannot alone validate model ordering.

The evidence includes incorrect polynomial values and roots in a retrieval task, inconsistent power calculations in an electromagnetic self-explanation task, and a linear rather than square-root pendulum relation in an emergent project.

Skill	Pairs	J_c MAE	J_r MAE	R MAE	R ρ	Direction matches
Adaptive hints	18	7.13	5.69	1.10	0.422	9/18
Backward design	18	3.37	3.39	1.45	0.756	14/18
Differentiation	18	12.83	17.22	2.95	0.423	11/18
Emergent projects	17	2.97	3.90	1.17	0.032	11/17
Hinge questions	18	15.48	3.89	4.26	0.197	5/18
Lesson design	18	4.33	6.13	2.26	-0.127	11/18
Motivation	18	9.89	7.15	1.50	0.267	11/18
Project briefs	18	3.52	4.06	1.50	-0.015	4/18
Retrieval practice	17	18.28	10.61	3.70	-0.244	5/17
Emotional literacy	18	9.32	11.43	1.55	-0.385	6/18
Self-efficacy	18	7.65	8.21	1.22	0.572	15/18
Self-explanation	18	11.53	11.69	2.33	0.309	9/18
Socratic questions	18	5.88	5.17	1.69	0.399	13/18
Spaced practice	17	8.51	5.30	1.19	0.612	9/17

Table 13: Complete 14-Skill human–Judge breakdown, two task clusters per row. MAEs are percentage points. Reference J_r is expanded over pairs; its reused human labels are not independent observations.

Task origin	Tasks	Pairs	Judge J_c	Human J_c	R MAE
Source scenarios	18	161	90.11	84.93	2.23
Constructed tasks	10	88	96.11	88.23	1.56

Table 14: Task-origin breakdown. J_c is on a 0–100 scale; R MAE is in percentage points. These coverage-sampled means do not replace the 305-task leaderboard.

Other disagreements concern feasibility: mandatory emotion disclosure despite stated privacy protections, or a delivery summary that does not contain the claimed artifact. Reviewers assess the provided text rather than unseen files and reduce related dimensions rather than erasing valid teaching structure. References are subject to the same checks and are not gold solutions. External factual verification is not exhaustive; this study supports an auditable, Skill-specific diagnosis of scoring behavior, not universal correctness guarantees.

D Cost Accounting Details

Following EduIllustrate's token-based generation accounting (Bi et al., 2026), we distinguish a bounded API microcost from a retained agent execution. Neither includes judging. Public online tariffs observed on 12 October 2026 are used for estimates, not as evidence of the account's bill.

Bounded API pilot. We fix three inventory tasks before requesting outputs: photosynthesis diagnostic-question construction, an eight-week retrieval schedule, and a Chinese teaching-method consultation. Each reachable Ark model receives the same educational instruction and task, without tools or Skill text. Four requests run concurrently, with a 4,096-token output cap; GLM Flash uses low reasoning effort, and the other models have thinking disabled. No temperature override is supplied. This is not a rerun of native OpenCode conditions. GLM-5.3 returns a model/access error on this ordinary endpoint and is not replaced by a different model.

All 12 generation requests return usage and the requested version identifier, but five reach the cap. Their combined estimated generation cost is CNY 0.6430; four short availability probes add approximately CNY 0.0005. No Judge requests are issued. Latency includes transport and queueing under concurrent execution; it is not a pure model-compute measurement. The DS Flash pilot uses the ordinary Ark version, whereas the main native DS Flash experiment used a third-party channel.

The fixed task identifiers are `hinge-question-designer__01`, `spaced-practice-scheduler__03`, and `lesson-builder__cn26_05`. The requested Ark identifiers are `glm-5-3-flash-260828`, `deepseek-v4-1-flash-260910`, `deepseek-v4-pro-ga-260813`, and `doubao-seed-2-1-pro-260915`. No access to the latter two models' native generation databases is inferred from their direct pilot responses.

Cache and reasoning accounting. For returned API input I_j, cached input K_j, and completion O_j, with respective CNY-per-million prices p_I, p_K, p_O, we use

$$C_j = \frac{(I_j - K_j)p_I + K_j p_K + O_j p_O}{10^6}. \quad (3)$$

For multiple recorded model calls, $C_{\text{task}} = \sum_j C_j$ before averaging tasks. Reasoning is already included in API completion usage. No implicit cache hits occur in this pilot. GLM Flash input/cache/output rates are 0.80/0.23/2.80; Pro rates are 9.00/0.30/27.00; Doubao rates are 6.00/1.20/30.00. Ark DS Flash peak rates are 2.00/0.04/8.00; idle rates are 1.00/0.02/4.00. This pilot completes during weekday Beijing peak hours (09–12 and 14–18), so its DS Flash costs use peak prices (Volcano Engine, 2026). Missing usage and unavailable prices remain unknown.

Retained native execution. For the same three tasks, both locally available native-model databases retain complete No/With message usage. We sum unique assistant messages in the root and descendant sessions, checking their answer bindings and message/step/session token agreement. OpenCode stores uncached input separately from cache reads and answer output separately from reasoning; we reconstruct API prompt/completion totals without double counting.

GLM Flash's mean recorded input rises from 5,758 to 14,998 tokens and estimated cost from CNY 0.01283 to 0.02062 with package access. DS Flash's With mean includes an eight-call consultation trace, illustrating how repeated context can dominate a trajectory. These small-sample observations are not population efficiency estimates or quality-adjusted rankings. The accounting covers each retained execution, not discarded earlier runs, unrecorded transport attempts, infrastructure costs, or Judge calls. Pro/Doubao generation databases are not available locally; their retained Judge traces are not substituted.

Model	Terminal/3	Input tokens	Output tokens	Time (s)	Est. cost (CNY)
GLM-5.3 Flash	3/3	485	2023	26.7	0.0061
DeepSeek V4.1 Flash	2/3	471	2704	21.7	0.0226
DeepSeek V4 Pro	1/3	470	3204	53.6	0.0907
Doubao Seed 2.1 Pro	1/3	519	3062	84.3	0.0950

Table 15: Direct-API pilot on three fixed tasks per model. Means include all requests, including capped outputs. Terminal indicates a nonempty natural stop, not verified correctness. Output includes reasoning; prices use the dated Ark tariffs.

Model	Condition	Calls	Input tokens	Output tokens	Cost (CNY)
GLM-5.3 Flash	No	1.00	5758	2936	0.0128
GLM-5.3 Flash	With	2.00	14998	3079	0.0206
DeepSeek V4.1 Flash	No	1.00	5970	3611	–
DeepSeek V4.1 Flash	With	4.33	41776	5190	–

Table 16: Three-task native execution supplement to Table 15, reconstructed without new generation. All model-call usage in each retained execution is aggregated before taking the task mean. Output includes reasoning; input includes caches. DS third-party prices are unknown, not replaced by Ark prices. This table does not estimate the full 305-task mean.